

Electric Power Steering (EPS) – HARA Test Document

Purpose: Test document for the HARA AI Assistant prototype. It is intentionally structured to test document extraction, retrieval, malfunction identification, HARA workflow, requirement drafting and traceability.

1. Item Definition

Item: Electric Power Steering (EPS)
System function: Provide steering assistance to the driver according to the driver's steering demand.
Vehicle context: Passenger vehicle with electrically assisted steering.
Operating conditions: Urban driving, highway driving, lane changing and low-speed maneuvering.

2. Functional Inputs and Outputs

Element	Description
Driver steering input	Driver applies steering torque to the steering wheel.
Torque/position sensing	Provides steering-related input information to the EPS controller.
EPS controller	Processes steering input and determines assist demand.
Electric motor	Generates steering assistance torque.
Vehicle steering mechanism	Transfers steering torque to the road wheels.

3. Candidate Malfunctions, Hazards and Hazardous Events

ID	Malfunction	Potential Hazard	Hazardous Event
M1	Loss of steering assistance	Reduced driver ability to control vehicle	Loss of steering assistance during highway driving may result in unintended vehicle
M2	Unintended steering assistance	Unexpected change in steering behavior	Unintended steering assistance during highway driving may cause an unexpected
M3	Incorrect steering assistance	Incorrect steering response	Incorrect assistance during a lane change may cause the vehicle to deviate from t
M4	Delayed steering assistance	Delayed vehicle response	Delayed assistance during an evasive maneuver may prevent the intended steering

4. S/E/C Assessment and Candidate ASIL

Important: These values are test inputs for the prototype workflow. They are not an authoritative ISO 26262 classification or a safety approval.

ID	Scenario	S	E	C	Candidate ASIL
M1	Highway driving at 100 km/h	S3	E4	C3	D
M2	Highway driving at 100 km/h	S3	E4	C3	D
M3	Lane change at highway speed	S3	E4	C3	D
M4	Evasive maneuver at highway speed	S3	E4	C3	D

5. Candidate Safety Goals

SG-01	Prevent unintended steering assistance from causing an unintended vehicle trajectory.
SG-02	Maintain or safely transition steering assistance so that loss of assistance does not lead to an unreasonable risk.
SG-03	Prevent incorrect steering assistance from causing an unintended vehicle path.
SG-04	Detect relevant steering-assistance faults and transition to a defined safe state where applicable.

6. Candidate Functional Safety Requirements (FSR)

FSR-01	The EPS shall detect loss of required steering-assistance capability.
FSR-02	The EPS shall detect implausible steering input or assistance information.
FSR-03	The EPS shall prevent an unintended steering-assistance command from being applied.
FSR-04	The EPS shall provide fault information to the vehicle safety mechanism when a relevant fault is detected.
FSR-05	The EPS shall transition to a defined safe operating condition following a safety-relevant fault, where required.

7. Traceability Example

Malfunction	Hazardous Event	Safety Goal	FSR
M1	HE-01: Loss of steering assistance on highway	SG-02	FSR-01, FSR-05
M2	HE-02: Unintended steering assistance on highway	SG-01	FSR-03, FSR-05
M3	HE-03: Incorrect assistance during lane change	SG-03	FSR-02, FSR-03
M4	HE-04: Delayed assistance during evasive maneuver	SG-02	FSR-01, FSR-05

Prototype note: This is a synthetic test document created specifically to exercise the HARA Assistant workflow. It is not a certified safety artifact and must not be used as an engineering approval document.